The Copy Ceiling: An Input-Exposure Control for Ontology-Grounded Generation over Private Corpora

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Abstract

When a language model answers over a private, curated corpus through graph-based retrieval, the reported "grounding uplift" hides a measurement problem: if the gold answers derive from the same corpus that is injected, uplift conflates faithful *delivery* of exposed facts with *reasoning* over injected structure. We introduce the *copy ceiling*, the recall a verbatim copy of the shown context would already achieve, and the signed *gain over copy*. The control is deterministic, judge-free and per-item, reported as a standing column beside generation recall, and paired with a placebo verified irrelevant by seed-disjointness; together they distinguish exposure coverage from beyond-exposure recovery before an architecture is chosen. On a production ontology-serving node, unaided models score ≈ 0.26 in-domain and grounded ≈ 0.92, yet across ten models from five providers the gain over copy is uniformly negative (-0.067 to -0.022). A direct exposure/recovery decomposition of 11,360 gold items shows why: three unexposed gold items are recovered in total, context utilisation spans 0.900–0.955, and models differ only in how much exposed gold they drop. The uplift is delivery, measured rather than inferred. A paired study on the same node, holding the model constant and varying only the serving path, lifts judged quality by $+0.27$ pooled ($p = 0.004$) and $+0.79$ where curation is deepest; a four-arm control study is consistent with content-specific transfer (the served block lifts quality, $+0.59$; the verified-irrelevant placebo's point estimate is near zero, $+0.04$, though differential attrition leaves the contrast imprecise). Out of domain the gated node does not regress: the general-set delta ($+0.05$, $[0.00, +0.13]$) lies inside a pre-specified ± 0.25 equivalence margin. Grounding evaluations whose gold derives from the injected corpus should report a copy ceiling as standard.

1 Introduction

A recurring enterprise requirement is a language model that answers accurately over a large, private, domain-specific corpus (a customer's product data, an organisation's internal knowledge, a curated scientific ontology) that the model cannot have memorised because it was never in pretraining. Retrieval-augmented generation is the standard remedy for this gap between parametric memory and the knowledge an answer requires [10, 23], and it helps most where parametric memory is weakest: long-tail and proprietary facts [29]. Structured, graph-based retrieval sharpens it further [3, 6, 14, 32]. But evaluating such a system faces a measurement problem the headline "grounding uplift" hides: when the gold answers are themselves derived from the corpus that is injected, a large raw-to-grounded gain conflates two very different behaviours, faithful *delivery* of the exposed facts and *reasoning* over the injected structure. Published knowledge-graph grounding reports the gain without a standing control that separates them.

We introduce the *copy ceiling*[1], a per-item, graph-derived input-exposure baseline, and the signed *gain over copy*: a deterministic, judge-free per-item exposure ceiling,

[1]The term *copy ceiling* is used independently and in an unrelated sense by Liu [27] (a decoding-layer editing mechanism, where it denotes token reachability under a line-level copy primitive; that paper was withdrawn in 2026), which we flag only to head off a terminology collision.

reported as a standing control beside generation recall and paired with a verified seed-disjoint placebo, that separates faithful delivery of injected facts from reasoning over ontological structure in graph-grounded generation. For each question the copy ceiling is the recall a no-op extractor of *exactly* the text the model was shown would already score; the gain over copy is the model's grounded recall minus that ceiling. It is the retrieval-augmented, per-item, continuous generalisation of the discrete input-only baselines long used to interpret reading comprehension [19] and natural-language inference [15, 36]. Its rhetorical role is a control the reader calibrates the headline number against, and the claim we draw from it is methodological: grounding evaluations whose gold is derived from the injected corpus should report a copy ceiling as standard [21, 30].

The setting makes the control load-bearing. Unaided, the models we test score ≈ 0.26 on in-domain questions (the domain is genuinely private, absent from parametric memory) and ≈ 0.92 given the curated context. Almost the whole of that uplift is exposure: across ten contemporary models from five providers the gain over copy is small and *uniformly negative* (-0.067 to -0.022), and beyond-exposure recovery, measured directly at item level, is negligible (three items in 11,360; §7.2), so models differ almost entirely in how much surfaced gold they drop. This is not a defect to explain away. High-fidelity delivery of an authoritative, human-vetted source *is* the product for private-knowledge grounding: the answer is trustworthy because the source is, attributable to a named generation, and cheap because the expensive curation is amortised once rather than re-paid per query.

Contributions.

1. **The protocol (C1, singular).** A deterministic per-item exposure reference (the copy ceiling and its signed gain over copy), the item-level exposure/recovery decomposition that interprets it, and a placebo verified irrelevant by construction, with a formal definition and an explicit assumptions list (§5). The signed quantity re-centres recall against exposure; the decomposition is what separates dropped exposed facts from beyond-exposure recovery.
2. **The evidence (C2).** Ten models are uniformly negative on gain over copy; a four-arm negative-control design (true / shuffled / masked / irrelevant scaffolds) probes whether the gain is content-specific; and a production-node paired study isolates the ecosystem feature by holding the model constant and varying only the serving path (§6; results in §7).
3. **The organising frame (C3).** A conjunctive, multivariate bar (in-domain delivery fidelity *and* out-of-domain non-regression, measured on two different instruments) within which the copy ceiling is the in-domain axis (§2, §9).
4. **The artefacts (C4).** An open harness (paired bootstrap, Wilcoxon, matched-pairs rank-biserial, Holm), the frozen question sets, all per-row data, and the served system itself.[2]

2 The Private-Corpus Setting

The design decisions this paper measures are not preferences; they are forced by a chain of deployment requirements. We state the chain so that the evaluation bar, and the system in §4, read as constraints rather than choices. Each requirement carries one design consequence and its architectural decision record (ADR); Table 1 records where each is enforced. The object of study is a served node[3] over a corpus built by a separate, CI-gated pipeline[4] and accelerated by an

[2]The Loom serving node: `https://github.com/DreamLab-AI/loom`.

[3]The served node `https://github.com/DreamLab-AI/loom` (code under the repository licence; corpus terms per the sibling knowledgeGraph repository).

[4]The canonical builder `https://github.com/DreamLab-AI/knowledgeGraph` (published at `https://narrativegoldmine.com`) runs `pytest pipeline/tests` and `pipeline.validate` before publishing each generation.

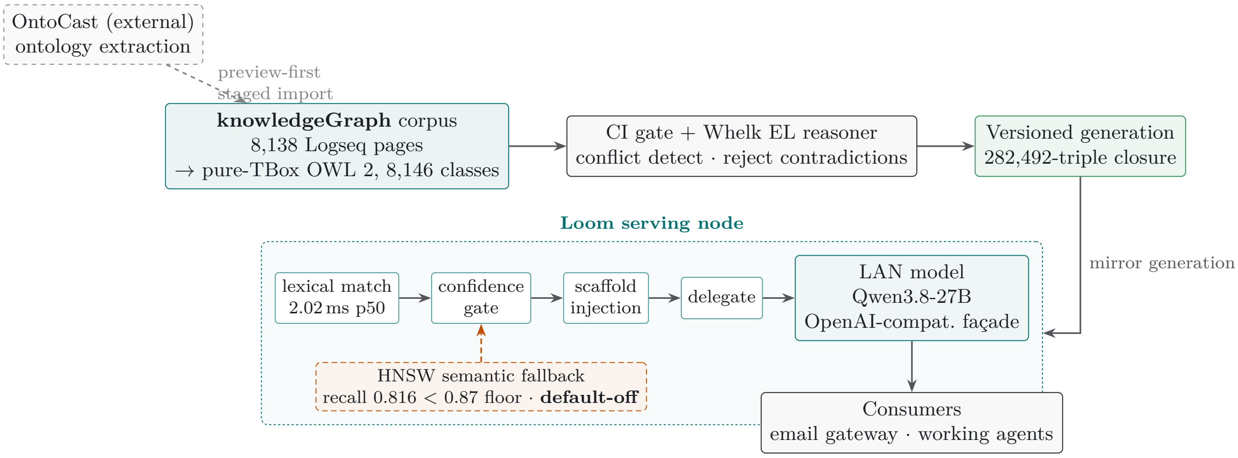

Figure 1: The data path around the measured node. Knowledge enters at the *knowledgeGraph* corpus (8,138 Logseq pages compiled losslessly into a pure-TBox OWL 2 ontology of 8,146 classes); output from OntoCast, an independent third-party extractor, enters only through a preview-first staged import that reports collisions and emits review candidates rather than publishing machine output directly. A CI gate and a Whelk OWL 2 EL reasoner classify each build and reject contradictions before publishing a versioned generation (a 282,492-triple reasoned closure). The *Loom* node mirrors a generation and serves it through a model-free pipeline (lexical match at 2.02 ms, confidence gate, scaffold injection) before delegating to whatever LAN model sits behind its OpenAI-compatible façade (Qwen3.8-27B here); consumers hold the façade URL and inherit grounding. The HNSW semantic fallback is drawn dashed in burnt orange because it ships default-off: its document-embedding recall (0.816) is below the 0.87 design floor, a red gate we report rather than hide (§2.1, §4).

in-process HNSW index[5].

2.1 The ecosystem in brief

This is a technical report on one component of a working stack (VisionFlow), and the component is easier to evaluate when the reader can see the whole data path. Knowledge enters at the corpus repository (*knowledgeGraph*): 8,138 Logseq markdown pages that compile losslessly into a pure-TBox OWL 2 ontology (8,146 classes, no individuals) published as versioned generations. Content arrives there two ways: curated authorship under human direction, and, recently, automated extraction. Output from OntoCast [13], an independently developed open-source ontology-extraction tool that is not our work, is staged through a preview-first import that reports identity collisions, refuses overwrites, and emits private review candidates rather than publishing machine output directly; the integration treats it as an upstream producer at a governed boundary, and neither project vendors the other. A CI gate (consistency tests plus a native conflict detector) and a Whelk OWL 2 EL reasoner [4] classify each build and reject contradictions before publication; the reasoned closure (282,492 triples) ships with the generation. The *Loom* node measured in this paper mirrors a generation and serves it: lexical retrieval over class titles, confidence-gated injection of per-IRI markdown blocks, read-only SPARQL over the closure, and delegation to whatever local model sits behind its OpenAI-compatible façade (at the time of writing, Qwen3.8-27B on two workstation GPUs). *RuVector* supplies the vector substrate: the corpus embeddings live in its Postgres store, and an in-process HNSW index is wired as a semantic fallback, currently disabled by default because its measured recall (0.816) is below the 0.87 design floor; a status we report rather than hide. Downstream, agent consumers (an email gateway, working agents) hold the façade URL and inherit grounding without integration work. Nothing in this description is load-bearing for the contribution; it exists so that §4–§7 can be read as measurements of a deployed system rather than a laboratory apparatus. Figure 1 draws the whole path.

[5]RuVector: `https://github.com/ruvnet/ruvector`.

1. **Private-knowledge need ⇒ curated, amortised corpus.** The model must answer over a corpus it cannot have memorised (raw ≈ 0.26 is consistent with a corpus absent from pretraining, not a proof of it); the answer is grounded in a human-directed, provenance-tracked ontology whose expensive curation is paid once and reused per query. The copy ceiling is the evaluation-time expression of "the curation, not the model, carries the answer" (ADR-135 D2; ADR-136 D4).

2. **Trust ⇒ human auditability at single-entity granularity.** A grounded answer is trustworthy only if a human can read and audit the served knowledge end-to-end; the canonical served unit is therefore one per-IRI markdown-with-ontology block (curated prose headed by its typed relations), and every retrieval port returns, or resolves to, an `Iri` addressing that unit (Invariant I-P1; ADR-136 D1).

3. **Auditability ⇒ accelerators strictly behind the markdown.** Nothing a machine indexes may become the thing a human trusts instead of the markdown; the lexical index, HNSW, oxigraph SPARQL and the attestation ledger are projections that find, rank or attest the unit and resolve back to its IRI, enforced by an acyclic crate ring rather than reviewer vigilance (ADR-136 D1–D3).

4. **Privacy / LAN-confinement ⇒ local delegation, no cloud on the hot path.** Content and answers must stay on the local network; generation is delegated to a LAN/local backend by URL and the augmentation hot path is fully in-process (ADR-135 D1; ADR-136 §3).

5. **Local-model churn ⇒ model swap without consumer change.** Local models change often (Gemma → Muse-Glimmer → Qwen3.8-27B); a stable OpenAI-compatible façade makes the backend one configuration line and lets model identity ride in the result, never the endpoint. That is exactly what lets this study hold grounding constant while varying the model under test (ADR-135 D1).

6. **Bounded latency ⇒ retrieval works with no model, fast.** Retrieval cannot depend on a model call: the scaffold, SPARQL and search endpoints need no model, and the lexical inverted index resolves in **2.02 ms** at the median over 8,146 class titles (ADR-136 §3).

7. **Injection interference ⇒ confidence-gated, benchmark-first injection.** Injected context can displace correct parametric knowledge on off-ontology queries [28, 39, 41]; a single injection authority scales the budget to the retrieval score and skips below a threshold [29], and features that could add noise are benchmark-gated (ADR-136 D3).

8. **Never-mixed provenance ⇒ attestation and boundary discipline.** Every grounded answer must trace to a named, versioned generation; each response carries a `loom` telemetry block and `corpusNature` honesty metadata, and the node serves only the published ontology, never the working graph (ADR-135 D6/D7; ADR-136 D5).

The bar these requirements imply is multivariate and conjunctive: excellent, attributable recall where the corpus is authoritative, *and* no regression on the general questions a model already answers. We introduce that framing here and reconcile the axes in §9.

3 Related Work

Knowledge-graph and private-corpus grounding. From the original RAG formulation [23] to graph-structured retrieval [6, 14], zero-shot knowledge-graph triple prompting [3], and the unifying roadmap and surveys [32, 35], the family grounds generation in retrieved graph content; GraphRAG motivates precisely our setting (answering over "private and/or previously unseen" collections [6]), and enterprise variants foreground the amortised-curation constraint [40]. Two features distinguish our object of study. Published work typically injects *instance* triples or opaque encodings: community summaries [6], say, or a GNN soft-prompt over a retrieved subgraph [16].

Table 1: The private-corpus requirements arc: each business requirement, the design consequence it forces, and the point at which the consequence is enforced (not merely intended). ADR = the governing architectural decision record.

Requirement	Design consequence	Enforcement point
Private, unmemorised corpus	Human-curated ontology, curation amortised once	Sha-addressable generation; raw ≈ 0.26 (ADR-135 D2)
Human auditability	Per-IRI markdown-with-ontology as the served unit	Ports resolve to `Iri` $\rightarrow$ `CanonicalUnit` (I-P1)
Accelerators behind the unit	Indexes only find/rank/attest the markdown	Acyclic crate ring; adapter cannot mint the unit
LAN confinement	Model delegated by URL; in-process hot path	`DISTILL_BACKEND_URL`; Profile A network-free
Model churn	Stable façade; model identity in the result	OpenAI-compatible `/v1`; one config line
Bounded latency	No-model retrieval path	Lexical index 2.02 ms p50; scaffold needs no model
Injection interference	Confidence-gated selective injection	Single injection authority; skip below threshold
Never-mixed provenance	Per-answer attestation; ontology-only boundary	`loom` block + `corpusNature`; working graph never served

We inject *schema-level* content (class definitions, typed relations, taxonomy) instead, and keep the served unit a human-readable markdown block; and none of these reports an input-exposure control, so their uplift is uncontrolled for what the injected context already contains.

Baselines that reframe a result. A recurring genre installs a simple control and overturns the naive reading of a headline: a strong, simple RAG baseline matches elaborate pipelines under matched token budgets [21]; a controlled comparison reverses a prior "RAG wins" finding and adds a confidence router [25]; apparent argument comprehension is shown to be spurious-cue exploitation [31]. Our paper belongs to this genre: the copy ceiling is the control, and "uplift is exposure, not reasoning" is the reframing. The confidence router of Li et al. [25] and selective retrieval more broadly [2, 18, 29] are the design precedent for our injection gate: they route retrieve-versus-not; we gate inject-versus-abstain.

Input-only baselines and faithfulness metrics. The intellectual root of the copy ceiling is the input-only baseline: passage-only / question-only baselines reveal that much apparent competence is shortcut exploitation of what the input already exposes [19], as do hypothesis-only baselines in natural-language inference [15, 36]. Where these are discrete, per-task classifiers, the copy ceiling is per-item and continuous, for graph-grounded generation. On the metric side, faithfulness is separately measurable from correctness [1, 8, 43], and metric-first evaluations give the metric a formal definition with an explicit assumptions list [30] and a shortcut-robustness subsection [9]. FActScore's precision-against-a-source and Adlakha's token-overlap K-Precision are cousins of our exposure count; ALCE treats "copy the passage" as a shortcut to *reject*, whereas we make that copy the metric's reference point. Contamination audits [12, 38] measure how much of a score is visible in the input post hoc; we make the deliberate exposure the reference a priori.

The nearest neighbours, and how we differ. The closest design precedent is a causal intervention that removes, substitutes or injects the gold answer to test whether answer presence drives retrieval-augmented gains [24]; we take only its paired-intervention *method* as precedent, and note that its authors withdrew it in 2026 after reporting errors that undermine its main conclusions, so we do not lean on its findings. A separate line builds leakage-free benchmarks by extracting a reasoning graph from question–context pairs and applying type-constrained entity replacement under a reasoning-graph consistency check and a leakage filter [26]; that operates at corpus-construction time, whereas the copy ceiling is a per-item measurement. On the metric side, RAGChecker decomposes RAG quality into claim-level retrieval and generation metrics by LLM-extracting and grading claims [37]; our context utilisation is, plainly, a deterministic surface-matched instantiation of RAGChecker's context-utilisation metric, and we claim no ownership of the decomposition idea. MIRAGE scores reader behaviour against oracle context across retriever–model pairings as benchmark-level adaptability [34], and the RAGAS *library*'s "context recall" (a later addition to the library, not a metric of the RAGAS paper [8]) is an LLM-judged, retrieval-side coverage number. What we have not found reported elsewhere is the specific combination: a deterministic, judge-free, per-item and continuous exposure ceiling on a schema-level ontology corpus, printed as a standing control beside the generation recall it bounds and paired with a verified seed-disjoint placebo. We keep the exposure and measure against it rather than removing it, and read the signed gain over copy *across models* as an exposure-normalised recall scale on which raw uplift cannot separate them. Judged arms use a cross-family judge to avoid self-preference [33, 42].

4 The Ontology Scaffold

The system measured here is the shipped serving node: a single static Rust binary organised as an eight-crate hexagonal workspace: a pure `loom-domain` core holding the port traits, five adapters, a `loom-scaffold` policy crate and a thin `axum` façade. The crate ring is the enforcement mechanism for the audit invariant of §2: only the domain crate mints a `CanonicalUnit`, so an adapter physically cannot return its own row, triple or vector as the served payload (Invariant I-P1). The node's retrieval, gating and scaffold serialisation are pinned byte-for-byte to a reference implementation by golden fixtures.

The canonical unit and the corpus. The served, reviewable unit is one per-IRI markdown-with-ontology block: curated research prose headed by its typed relations (`subClassOf`, `requires`, `enables`, `uses`, `relatedTo`, `contrastsWith`). The corpus is one sha-addressable *generation* (**8,146 concept classes** and **282,492 triples** in the Whelk-reasoned closure at the generation evaluated here), mirrored atomically so the node always serves a known, mixed-build-free generation; it is not baked into the binary. We explicitly reject the trajectory in which knowledge degrades into opaque community summaries or GNN-encoded subgraphs [6, 16]; the reviewable markdown is the durable asset, and the accelerators below only find, rank or attest it [11].

The lexical matcher and the confidence gate are the *sole* injection authority: every candidate flows through one policy that scales the scaffold budget to the retrieval score and injects nothing below a threshold, the mechanism by which the node avoids "going jagged" on off-ontology queries (§7.5). The graph store answers read-only, LIMIT-clamped SPARQL over the reasoned closure with no model call. An in-process HNSW semantic fallback over 384-dimensional embeddings is wired for out-of-vocabulary queries but ships *default-off*: its measured document-embedding recall (0.816) is below the 0.87 design floor, so it stays disabled pending a query-shaped embedding that clears a multivariate benchmark rather than a threshold relaxation. We return to this honest red gate in §10. Finally the model-swap seam: generation is delegated to whatever backend `DISTILL_BACKEND_URL` names (Qwen3.8-27B today), model identity rides in the response, and swapping the model changes no consumer, which is what lets §6 hold grounding constant and vary the model, or hold the model constant and vary the serving path.

Table 2: The request pipeline. Retrieval, gating and the scaffold serialisation need no model; only the final delegation calls one. The single injection authority (match + gate) is the sole path by which any candidate reaches the model, whether lexical or, when enabled, semantic.

Stage	Mechanism	Model?
Match	Lexical inverted index over 8,146 class titles; 2.02 ms p50 → seed classes	no
Gate	Single injection authority: budget scaled to retrieval score, skip below threshold	no
Scaffold	Budget-clamped serialisation of the matched units' markdown-with-ontology	no
SPARQL / search	Read-only, prologue-aware LIMIT-clamped oxigraph over the reasoned closure	no
Semantic fallback	In-process HNSW over 384-d embeddings; fires on lexical miss → candidate seeds into the same gate	*default-off*
Delegate	Inject scaffold into the last user message; call `DISTILL_BACKEND_URL`	yes

5 The Copy Ceiling

The copy ceiling is the paper's primary contribution, so we give it a formal definition, an explicit assumptions list, and a robustness design of its own, following the metric-first precedent [9, 30].

5.1 Definition and assumptions

Fix a question q_i with gold target set G_i (the {slug, title} pairs the graph asserts as its answer), the exact scaffold text S_i shown to the model, and the model's answer a_i. Let $\mathrm{m}(t, X) \in \{0, 1\}$ be the deterministic surface matcher: $\mathrm{m}(t, X) = 1$ when the normalised title of gold item t, or at least 80% of its length-≥ 4 words, appears in text X. Define the exposed count and the per-item **copy ceiling**

$$e_i = \sum_{t \in G_i} \mathrm{m}(t, S_i), \qquad c_i = \frac{e_i}{|G_i|},$$

and the grounded recall $r_i = (\sum_{t \in G_i} \mathrm{m}(t, a_i))/|G_i|$. The copy ceiling is the recall of the no-op extractor that returns S_i verbatim as its answer: that answer scores exactly c_i by construction. For the raw arm S_i is empty, so $c_i \approx 0$.

Assumptions (when the instrument applies).

(A1) **Deliberate exposure.** Questions, gold and scaffold derive from one graph, so the gold is present in the injected text by design. The ceiling *measures* this exposure; it is not a leak to remove.

(A2) **Shared matcher.** c_i and r_i use the same matcher, so both are lower bounds under paraphrase and their difference is first-order insensitive to its under-counting. The cancellation is only approximate: the matcher's error rate differs between the structured scaffold text scored for c_i and the free generated prose scored for r_i, so a residual asymmetry survives; where paraphrase is a live threat we re-score with a semantic judge (§6).

(A3) **Per-item, model-free ceiling.** c_i is a property of (q_i, S_i) only; for a fixed scaffold it is identical across models, which is what makes gain over copy comparable across a model sweep.

(A4) **Graph-derived gold.** The ceiling is informative only when gold is derived from the injected corpus. Where gold is authored independently of the scaffold (the out-of-domain arm), $c_i \approx 0$ and a different instrument, judged quality, carries the axis.

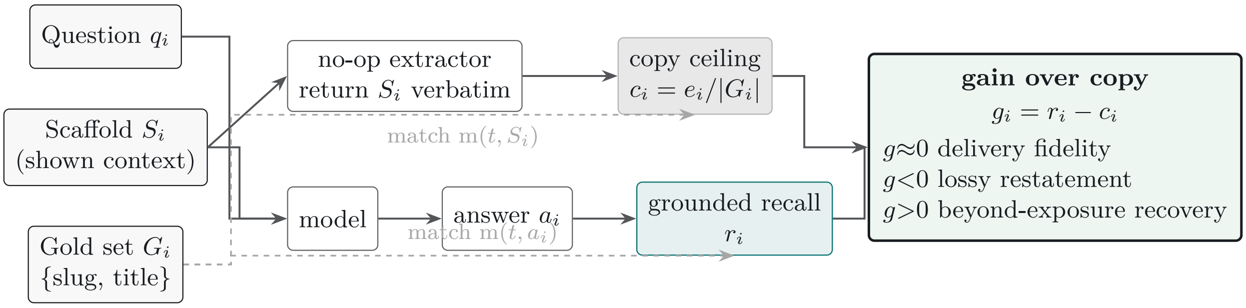

Figure 2: The copy ceiling and the signed gain over copy on one panel. The same shown context S_i feeds two extractors: a no-op extractor that returns S_i verbatim, whose recall against the gold set G_i is the per-item *copy ceiling* c_i; and the model, whose answer a_i scores the grounded recall r_i. Both scores use the identical surface matcher $\mathrm{m}(t, \cdot)$ with G_i as reference (dashed), so their difference, the *gain over copy* $g_i = r_i - c_i$, is first-order insensitive to the matcher's under-counting. The sign reads directly: $g \approx 0$ is faithful delivery (the target behaviour for private-knowledge grounding), $g < 0$ is lossy restatement of surfaced gold, and $g > 0$ would evidence recovery of gold beyond what the exposure check finds (§5).

A recall reference point, not an endorsement. The no-op extractor that defines the ceiling scores c_i by construction, but it is not an admissible answer: it has no response precision, conciseness or relevance, and returning the whole scaffold verbatim would fail all three. The copy ceiling is therefore a recall reference point against which delivery fidelity is read, not a recommendation to dump context into the answer.

5.2 Gain over copy

The signed **gain over copy** is $g_i = r_i - c_i$, with headline $\bar{g} = \bar{r} - \bar{c}$. It re-centres grounded recall against the exposure reference and reads directly:

- $g \approx 0$, *delivery fidelity*: the model restates nearly everything it is shown. Recall measures omission, not fabrication, so what the model adds *beyond* the exposed gold is unmeasured here; for private-knowledge grounding this restatement is the target behaviour, not a shortfall.
- $g < 0$, *lossy restatement*: the model drops surfaced gold, a copy-fidelity failure. The decomposition of §7.2 measures this directly: the negative gain is almost entirely exposed-but-omitted gold (n_{10}), with unexposed recovery (n_{01}) negligible. The magnitude ranks models where near-identical raw uplift cannot.
- $g > 0$ would evidence *reasoning over structure*: recovering gold the exposure check missed, for example by composing a relation the scaffold implies but does not state. We observe essentially none in-domain ($n_{01} = 3$ of 11,360; §7.2). A positive g would not by itself prove reasoning: it could equally be parametric leakage, guessing, or matcher asymmetry between scaffold and answer, and only the multi-hop design of §10 could separate them.

A near-ceiling result therefore licenses "faithful delivery" and forbids the inference "reasoning over ontological structure". This is the distinction the rest of the paper turns on; Figure 2 draws the two paths the instrument compares.

5.3 Negative controls: what would move the gain

A single objection can sink the instrument: that a positive-looking delivery is lexical overlap between answer and scaffold, not knowledge transfer. We answer it with four arms that hold the model and the question fixed and perturb only the injected block (defined here; results in §7.4):

true the block the node actually served, injected into the bare model. It should reproduce the live loom arm, proving the serving path adds nothing beyond the block itself.

shuffled the same block with its body sentences shuffled (wrapper kept, seeded). Same tokens, destroyed sentence order: if the gain survives, it was lexical soup. This is a weak probe of relational structure, since shuffling whole sentences preserves the facts *within* each sentence.

masked the same block with every seed-class title replaced by `Entity-`k. Structure intact, entity names gone: if the gain survives, the score was matching names, not knowledge. Because only seed-class titles are masked, answer-bearing target names remain, so this is a weak probe of name-matching.

irrelevant the well-formed, on-corpus scaffold of a *different* question (a fixed derangement over the engaged items). If the gain survives, any ontology-shaped text would do.

The logic is differential. If gain over copy stays near zero under **shuffled**, **masked** and **irrelevant**, the ceiling is tracking exposure faithfully and the delivery is content-bound; if it moves, the ceiling was leaking through surface form or position. This is the experiment that tests whether the observed lift is content-specific, and it is the direct analogue of a shortcut-robustness check for a faithfulness metric [9].

6 Method

Design. Both studies are paired within-model: every comparison is between two answers to the *same* question by the *same* model, so between-question difficulty variance cancels and the estimand is a paired per-question difference. The two studies isolate different variables: the ten-model sweep varies the *model* with grounding fixed; the new study varies the *serving path* with the model fixed.

Study 1: the ten-model sweep. Using the knowledge graph as its own oracle we template three question types from every sufficiently-defined class: **T-REL** (what a class requires / uses / depends on / enables; its parts), **T-TAX** (immediate and ancestral parents) and **T-COMMON** (a pair's shared ancestor), with gold the {slug, title} targets the graph asserts. Seed 42 yields **510 questions** over 15 domains (285 T-REL, 180 T-TAX, 45 T-COMMON). Each is asked *raw* and *scaffold* (a 1,500-token budget-clamped ontology extract prepended), across ten models from five providers under one deterministic configuration (`temperature`=0, `max_tokens`=2048; `reasoning_effort`=low only where thinking cannot be disabled, so a mandatory pass does not truncate the answer). Scoring is the lexical matcher of §5: a gold item is a hit when its normalised title, or $\geq 80\%$ of its long words, appears in the answer. This under-counts paraphrase, so absolute recall is a lower bound and we treat the *paired* delta (same question, scorer and model) as the signal, flagged honestly as a lexical estimand. Transport-level truncations were detected per row; GLM-4.6 truncated on 174 rows, of which 5 were retried and the remainder scored as received, so that row is treated as lower-confidence (Gemini 2.5 Flash-Lite truncated on 16).

Study 2: the production-node paired study. To isolate the ecosystem feature rather than the model, we hold the model constant (Qwen3.8-27B on the reference GPU host) and vary only the serving path. The *loom* arm calls the production Rust node's `/v1/chat/completions`, which retrieves, confidence-gates and injects the scaffold and returns per-answer telemetry (engagement, injected tokens, fusion path) in the response `loom` block; the *raw* arm calls the same model backend directly with no scaffold. Both use `temperature`=0 and `max_tokens`=1536, below which the reasoning backend truncates to empty. Questions are three frozen sets of **117** questions in total (24 arcane, 33 thin, 60 general): *arcane* (deep in-domain), *thin* (sparsely-covered) and *general* (out-of-domain), frozen before any Study 2 generation; the general set is the set graded by the out-of-domain judged arm below, so the two judged arms are comparable. The four negative-control arms of §5.3 run against the same node: scaffolds are fetched from the LLM-free `/loom/scaffold` endpoint, so every control injects exactly the block the loom arm served, and presented to the bare model as a system message.

Out-of-domain judged arm. Complementing the lexical sweep, a five-model judged arm asks the 60-question *general* set (adjacent, in-domain-general and off-domain subsets) *harnessed* and *bare* and grades each pair 0–5 against frontier-authored key points, giving the out-of-domain axis an instrument of its own (§7.5).

Cross-family semantic judge. Because Study 2 spans in-domain and out-of-domain questions and must survive the paraphrase objection, answers are scored by a reference-guided **0–5 rubric** judge rather than lexically. The judge is `openai/gpt-4.1` served via OpenRouter at `temperature`=0, `max_tokens`=200, under a fixed system-message rubric identical across every arm; it is a different model family from the Qwen candidate to avoid self-preference [33, 42], blind to which arm and which study each answer came from, and the grading job is resumable per $(\mathrm{set}, \mathrm{id}, \mathrm{arm})$. The gold references are the graph-derived targets in-domain and frontier-authored key points out-of-domain.

Statistics. The headline per set and pooled is the paired mean difference with a seeded **10,000-resample bootstrap** 95% percentile interval [7, 20], complemented by a two-sided **Wilcoxon signed-rank** test and, as the paired effect size, the **matched-pairs rank-biserial correlation** $r = (T^+ - T^-)/(T^+ + T^-)$ over the non-tied pairs. We use the rank-biserial rather than Cliff's delta because the design is matched pairs: Cliff's delta is an independent-samples dominance statistic and would divide the win−loss margin by the many tied pairs, whereas the rank-biserial conditions on the non-tied pairs. Because the three sets are reported together, set-level p-values carry a **Holm–Bonferroni** correction [5]. Where the claim is *equivalence* rather than difference, the out-of-domain non-regression axis, we use two one-sided tests (TOST) against a pre-specified margin [22] of ± 0.25, rather than reading a non-significant difference as no effect. In-domain, graph-derived questions cluster by class and domain [5]; we therefore report a domain-clustered block bootstrap alongside the naive interval and an intention-to-treat delta that retains questions on which the gate did not engage [17].

7 Results

Reporting deviations (Study 2). Two departures from the pre-specified plan shape how the production-node numbers are read, and we state them up front. *(i) Budget-exhaustion re-runs.* At the `max_tokens`=1536 budget, 42 of the 234 loom/raw completions (117 questions × two arms) spent the whole budget inside the reasoning trace and returned empty. Every affected pair was re-run at 4096 in *both* arms so the pairing is preserved; three questions were still empty at 4096 and are dropped, leaving the complete-case $n = 114$ paired analysis (the thin set falls from 33 to 30). Because the 39 surviving re-run pairs are compared at 4096 while the rest sit at 1536, the pooled contrast is a mixed-budget estimand over those pairs, which we flag rather than hide. *(ii) Controls not retried.* The four negative-control arms were not re-run at the larger budget, so at 1536 they carry substantial empty attrition: true 24/56 (42.9%), shuffled 20/56 (35.7%), masked 23/56 (41.1%), irrelevant 28/56 (50.0%). Every control contrast is therefore complete-case over the surviving pairs, with the small, unequal n this implies. The attrition is not noise to discard but a finding in itself: at a fixed reasoning budget the empty rate rises with scaffold length and disorder (the live loom arm's original-1536 empties were 2/24 arcane and 19/33 thin, against raw 6/24 and 15/33), a scaffold-length × reasoning-budget interaction we return to in §10. All control results below are read as lower-confidence, complete-case contrasts for this reason.

7.1 In-domain: delivery fidelity

On Gemini 3.7 Flash the paired design gives a large, tight uplift: raw recall 0.375, scaffold recall 0.942, paired delta **+0.567**, naive 95% CI [+0.532, +0.603], domain-clustered 95% CI [+0.524, +0.611] over $n = 510$ with zero errors and zero truncated responses (Figure 3). The low raw score is consistent with a genuinely private domain: parametric memory answers barely a third, as absence from pretraining would predict. The copy ceiling then *characterises* the grounded score: the mean

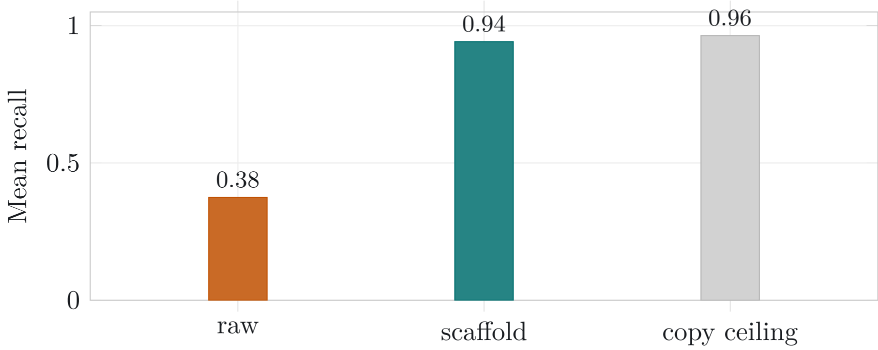

Figure 3: Gemini 3.7 Flash, $n = 510$. The scaffold arm (0.942) lands just under the *copy ceiling* (0.964), the recall a no-op extractor of the injected context would already achieve. The +0.57 raw→scaffold uplift is almost entirely answer exposure, not reasoning over structure.

fraction of gold already present in the injected scaffold is **0.964**, and the model's recall (0.942) tracks it within −0.022. The system does exactly what a private-knowledge grounding layer should: the curated ontology carries the answers and the model delivers them back with high fidelity. The tiny negative gain says the model restates nearly everything it is shown; recall measures omission, not fabrication, so what it adds beyond the gold is unmeasured here. What the ceiling licenses is high-recall delivery of the exposed targets (attribution precision is itself an unmeasured axis, §9), not free-association from parametric memory. Whether a model does more than restate (recovering gold the lexical check missed, or conversely *dropping* surfaced facts) is the axis §7.2 turns on.

7.2 Ten models: gain over copy

Table 3 runs the identical paired design across ten models from five providers under one deterministic configuration. Every model shows a large raw→scaffold uplift (+0.57 to +0.78) and every model lands just under the copy ceiling: scaffold recall spans 0.897–0.942 against a ceiling of **0.964** that is a property of the questions and the injected context (not the model), and is therefore identical across the row. The raw uplift is a near-constant restatement of exposed gold; it does not separate the models.

The signed **gain over copy** does. It is negative for all ten (Figure 4), from −0.067 (Gemini 2.5 Flash-Lite) to −0.022 (Gemini 3.7 Flash): on this lexical metric beyond-exposure recovery is negligible (the decomposition below counts 3 unexposed items recovered out of 11,360), so the models differ almost entirely in how much surfaced gold they *drop*. Because the ceiling is a single constant (0.964) across the row, ranking by gain is arithmetically the same as ranking by scaffold recall ($g_m = R_m - 0.964$); the ceiling introduces no new ordering. What it introduces is a *scale*: reading recall against a copy of the context rather than against the bare model turns ten near-identical raw uplifts into an interpretable fidelity spread, on which the most recent models (Gemini 3.7 and 3.5 Flash, Claude Haiku 4.5) sit closest to the ceiling and older or smaller ones (Gemini 2.5 Flash-Lite, GPT-4.1-mini, Mistral-Small-24B) fall further below it; the ordering is specific to this task, not a general capability ranking. Two models carried transport-level truncations under the fixed 2048-token budget (GLM-4.6: 174 truncated, 5 retried, the remainder scored as received and treated as a lower-confidence row; Gemini 2.5 Flash-Lite: 16 truncated), recorded per row rather than hidden.

What adds information: the exposure/recovery decomposition. The sign of g cannot on its own establish that models do not reason beyond exposure: a negative g is consistent with omitting exposed gold, recovering none of the unexposed gold, or both. We therefore score every gold item under a 2×2 contingency by whether the scaffold *exposed* it and whether the answer *recovered* it. Pooled over the ten models and 11,360 gold items the counts are $n_{11} = 9{,}865$ exposed-and-recovered, $n_{10} = 735$ exposed-and-omitted, $n_{01} = 3$ unexposed-and-recovered, and $n_{00} = 757$

Table 3: Ten models, one deterministic configuration, $n = 510$ (seed 42), sorted by gain over copy (scaffold recall − copy ceiling). The copy ceiling (0.964) is a property of the questions and the injected context and is identical across models. The interval shown is the domain-clustered block bootstrap (the wider, primary one).

Model	Raw	Scaffold	Uplift [clustered 95% CI]	Ceiling	Gain
Gemini 3.7 Flash	0.375	0.942	+0.567 [+0.524, +0.611]	0.964	−0.022
Gemini 3.5 Flash-Lite	0.323	0.938	+0.615 [+0.565, +0.667]	0.964	−0.026
Claude Haiku 4.5	0.151	0.934	+0.783 [+0.741, +0.827]	0.964	−0.031
GLM-4.6	0.243	0.928	+0.685 [+0.641, +0.727]	0.964	−0.036
DeepSeek-Chat	0.347	0.924	+0.578 [+0.532, +0.622]	0.964	−0.040
Llama-3.3-70B	0.227	0.916	+0.688 [+0.632, +0.738]	0.964	−0.049
Qwen-2.5-72B	0.309	0.914	+0.605 [+0.562, +0.649]	0.964	−0.050
Mistral-Small-24B	0.285	0.906	+0.621 [+0.558, +0.682]	0.964	−0.058
GPT-4.1-mini	0.162	0.905	+0.743 [+0.697, +0.787]	0.964	−0.060
Gemini 2.5 Flash-Lite	0.227	0.897	+0.670 [+0.623, +0.716]	0.964	−0.067

Table 4: Exposure/recovery decomposition over the 510-question sweep (1,136 gold items per model; 11,360 pooled), sorted by context utilisation. Utilisation is $n_{11}/(n_{11} + n_{10})$, the fraction of *exposed* gold the answer recovers; n_{10} is exposed gold the answer omitted; n_{01} is gold recovered though the scaffold did not expose it. Across all ten models n_{01} never exceeds 1, and is 3 pooled: recovery beyond exposure is empirically negligible. Counts are pooled over gold items (micro-averaged), whereas the headline recall and ceiling average per-question ratios (macro-averaged), so these counts do not reconstruct those numbers directly.

Model	Context utilisation	n_{10} (exposed, omitted)	n_{01} (unexposed, recovered)
Gemini 3.7 Flash	0.955	48	0
Gemini 3.5 Flash-Lite	0.947	56	0
Claude Haiku 4.5	0.946	57	1
GLM-4.6	0.945	58	0
Llama-3.3-70B	0.934	70	0
DeepSeek-Chat	0.931	73	0
Qwen-2.5-72B	0.924	81	1
Mistral-Small-24B	0.917	88	1
GPT-4.1-mini	0.908	98	0
Gemini 2.5 Flash-Lite	0.900	106	0
Pooled	**0.931**	**735**	**3**

unexposed-and-omitted (Table 4). The headline is n_{01}: only **3 of 11,360** gold items (0.026%) are recovered without being exposed, and no single model exceeds $n_{01} = 1$. "Models do not recover gold beyond what the scaffold exposes" is thus a direct item-level measurement, not an inference from the sign of g. The negative gain is then almost pure n_{10}: with $n_{01} \approx 0$, $g \approx -n_{10}/|G|$, so the shortfall under the ceiling is imperfect *copying* of exposed gold (lexical-match under-count of paraphrase included), a copy-fidelity deficit rather than evidence about reasoning in either direction. Per-model *context utilisation* $n_{11}/(n_{11} + n_{10})$ runs 0.900–0.955: even with the fact in front of it a model drops about one exposed item in fourteen. Utilisation is close to, but not identical with, the recall ranking. It pools over exposed items whereas headline recall averages per-question ratios and the ceiling c_i varies per item, so utilisation is *not* recall shifted by a constant at item level; the two orderings agree everywhere except one adjacent swap (llama-3.3-70b utilises exposed context marginally better than deepseek-chat, 0.934 versus 0.931, yet scores lower headline recall, 0.916 versus 0.924). The divergence is that single transposition, so "uses the context best" and "scores highest" nearly, but not exactly, coincide.

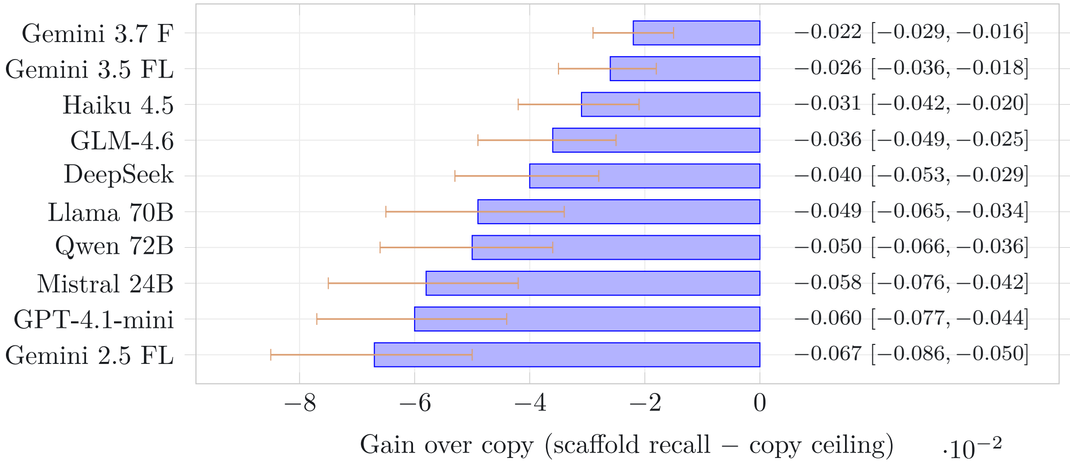

Figure 4: Signed gain over copy for all ten models, negative throughout, with per-model seeded 10,000-resample bootstrap 95% intervals (whiskers); the value column at right gives each point and interval. No model's interval crosses zero, and none exceeds the copy ceiling on the lexical metric; the spread (−0.067 to −0.022) is the axis that separates them where the near-identical raw uplift does not. Models nearest zero (top) restate the vetted facts most faithfully.

7.3 The production-node paired study

The ten-model sweep varies the model with a fixed offline scaffold; the deployment question is whether the *shipped* node (live retrieval, confidence gate, injection authority and all) moves answer quality when the model is held constant. Study 2 answers it: Qwen3.8-27B on the reference host, identical decoding, the only difference being whether the request passes through the Loom node (*loom*) or hits the backend directly (*raw*), scored 0–5 by the cross-family GPT judge of §6. Table 5 reports the paired result per set and pooled.

Pooled over $n = 114$, the loom path lifts judged quality by $+\mathbf{0.27}$ points ($[+0.11, +0.45]$, Wilcoxon $p = 0.0036$, matched-pairs rank-biserial $r = +0.589$), winning 24 pairs, losing 8 and tying 82. The effect concentrates where the corpus is deep: on the *arcane* set the lift is $+\mathbf{0.79}$ ($[+0.17, +1.42]$, $p = 0.033$, Holm-adjusted $p = 0.099$, $r = +0.670$); on the sparsely-covered *thin* set it is a positive but non-significant $+0.30$ ($[-0.03, +0.63]$); and on the out-of-domain *general* set it is a negligible $+0.05$ whose interval $[0.00, +0.13]$ sits entirely inside the pre-specified ± 0.25 equivalence margin: non-regression established by CI-inclusion, not merely by a non-significant difference (the TOST reading; §7.5). The point estimates (large where curation is deep, vanishing where the corpus is thin or the question is off-domain) are consistent with the hypothesised signature of a grounding effect, and run opposite to what a uniform prompt-formatting artefact would produce. Figure 5 shows the gradient and the equivalence band together.

The telemetry corroborates that the gate behaves as designed. Injection engagement is 95.8% on arcane, 100% on thin, and 78.3% on general. The general-set null is therefore the stronger reading it first appears: the scaffold was still injected on more than three-quarters of off-domain questions, and judged quality did not regress even so; the gate's declining to inject on the remaining 21.7% is a secondary mechanism, not the main explanation. Median injected context is $\approx$ 1,200–1,300 tokens across sets. Latency is at parity on the sets that matter for the claim: arcane medians 89.8 s (loom) versus 93.4 s (raw), general 66.0 versus 65.7 s; the thin set alone runs slower under loom (172.4 versus 145.2 s), consistent with its full engagement and longer scaffolds, and we report it rather than average it away.

7.4 Negative controls: attributing the gain

The live lift and the ten-model delivery both invite one objection: that the gain is lexical overlap between answer and injected block, not knowledge transfer. The four control arms of §5.3 test it directly, each holding the model and question fixed and perturbing only the injected scaffold, fetched

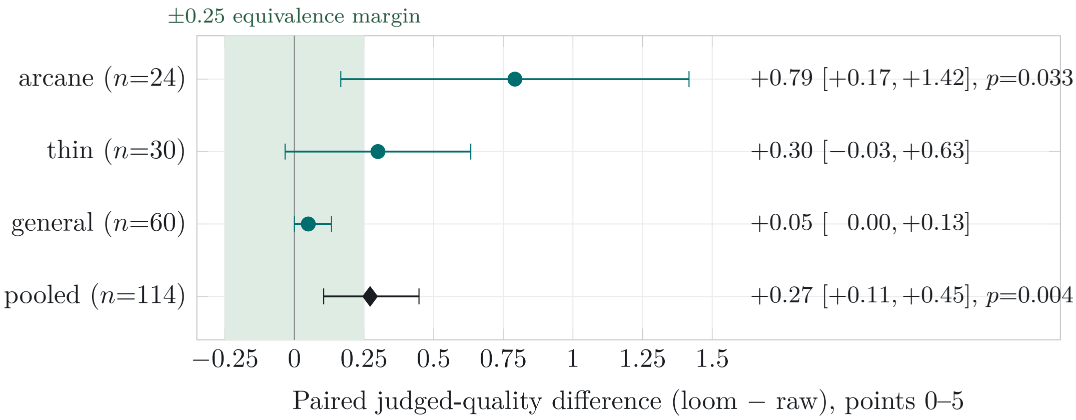

Figure 5: Study 2 paired judged-quality deltas (Qwen3.8-27B, model held constant, loom versus raw serving path; cross-family GPT judge), per set and pooled, with seeded 10,000-resample bootstrap 95% intervals. The three sets are ordered by curation depth: the effect is largest on the deep *arcane* set (+0.79), attenuates on the sparsely-covered *thin* set (+0.30), and is negligible out-of-domain on *general* (+0.05). That monotone gradient, large where curation is deep and vanishing where the corpus is thin or the question off-domain, is the signature a genuine grounding effect should show and not what a uniform prompt-formatting artefact would produce. The shaded band is the pre-specified ±0.25 TOST equivalence margin: the general-set interval $[0.00, +0.13]$ sits wholly inside it, so out-of-domain non-regression is read by CI-inclusion, not from a non-significant difference (§7.3, §7.5).

Table 5: Study 2, the production-node paired study: Qwen3.8-27B, model held constant, loom versus raw serving path, judged 0–5 by a cross-family GPT judge. Paired mean difference with seeded 10,000-resample bootstrap 95% CI, two-sided Wilcoxon p (Holm-adjusted across the three sets), the matched-pairs rank-biserial r (paired effect size, conditioned on non-tied pairs), and win/loss/tie counts. Complete-case after budget-exhaustion re-runs (§7).

Set	n	Δ (loom−raw)	95% CI	p (Holm)	r	W/L/T
arcane	24	+0.79	[+0.17, +1.42]	0.033 (0.099)	+0.670	10/3/11
thin	30	+0.30	[−0.03, +0.63]	0.118 (0.237)	+0.431	12/5/13
general	60	+0.05	[0.00, +0.13]	0.180 (0.237)	+1.000†	2/0/58
pooled	114	+0.27	[+0.11, +0.45]	**0.0036**	+0.589	24/8/82

† The general-set r rests on 2 non-tied pairs (58 ties); read it as directional only, not as a magnitude.

verbatim from the LLM-free `/loom/scaffold` endpoint. Table 6 reports the pooled (arcane + thin) contrasts; we read them as lower-confidence, complete-case comparisons for the attrition reasons of §7; Figure 6 plots the ladder of contrasts.

Three results triangulate the attribution. First, a **sanity check**: serving the exact block through the bare model (*true*) reproduces the live *loom* arm (loom − true is −0.06 [−0.38, +0.25], indistinguishable from zero), confirming the serving path adds nothing beyond the block it injects. Second, the **content effect is real**: true − raw is +0.59 [+0.16, +1.06] ($p = 0.026$). Injecting the served block lifts judged quality well above the bare model. Third, the **placebo's point estimate collapses**: an equally well-formed, on-corpus scaffold drawn from a *different* question (*irrelevant*) shows irrelevant − raw +0.04 [−0.54, +0.57] ($p = 0.73$). The near-zero point estimate is the qualitative anchor of the four, suggesting the lift is not a generic effect of ontology-shaped text; but the interval is wide under attrition, so the placebo is verified irrelevant by construction, not demonstrated empirically inert by equivalence. Subtracting the placebo from the content arm isolates the content-specific component, true − irrelevant = +0.35 [0.00, +0.78] ($p = 0.091$): a residual consistent with content-specific transfer, suggestive rather than established at conventional significance given the wide interval and $n = 23$ surviving pairs.

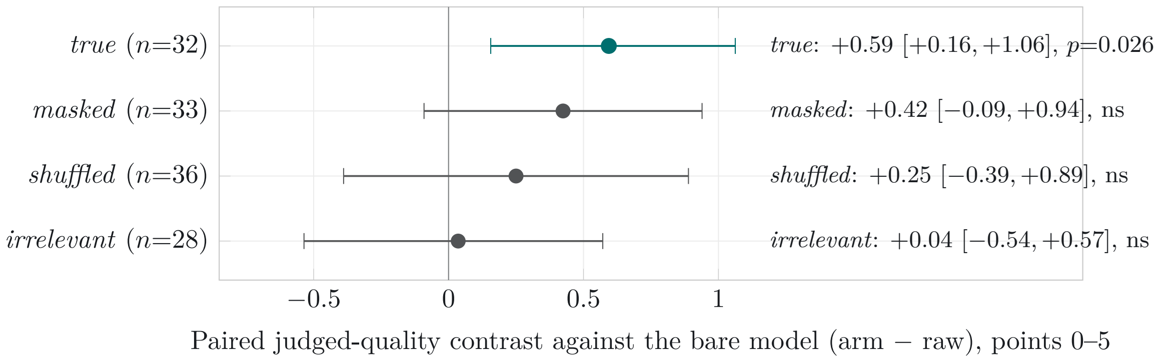

Figure 6: Negative-control decomposition (Study 2, pooled arcane + thin, judged 0–5 by the cross-family GPT judge), each arm holding the model and question fixed and perturbing only the injected block, read as a paired contrast against the bare model (arm − raw) with seeded bootstrap 95% intervals. Read top to bottom the ladder is the paper's central negative control: the served block lifts quality (*true* − raw +0.59, the one significant contrast, in teal), masking the seed-class names (+0.42) or shuffling the body sentences (+0.25) costs comparatively little, and an equally well-formed, on-corpus scaffold drawn from a *different* question shows a near-zero point estimate (*irrelevant* − raw +0.04). That near-zero placebo is consistent with the lift being specific to scaffold *content* rather than the mere presence of ontology-shaped text, though its wide interval under attrition means inertness is not demonstrated by equivalence; the content-specific residual is *true* − *irrelevant* = +0.35 [0.00, +0.78]. The arms are complete-case over unequal survivor sets and all but *true* are non-significant with wide intervals, so the attribution rests on the pattern across arms, not any single contrast (§7.4).

The placebo had to be verified irrelevant. This conclusion depends entirely on the irrelevant arm being a true placebo, and our first attempt was not. Constructing the "different question" donor by a shift-by-one derangement over the engaged items left donors in the *same* domain cluster as their targets, so the "irrelevant" scaffold routinely shared vocabulary and neighbouring concepts with the gold, and behaved like a weak content arm rather than a placebo. We rebuilt the derangement under a seed-IRI-disjointness constraint (every donor scaffold verified to share no seed IRI, and thus no seed class, with its target's gold) and only then did the arm's point estimate collapse to ≈ 0. A placebo assumed irrelevant rather than verified irrelevant silently inflates the very baseline it is meant to control; the fix was cheap and the lesson is general.

Masking and shuffling lose comparatively little. Replacing every seed-class *name* with `Entity-k` (*masked*) still beats raw by +0.42 (ns, wide CI), and shuffling the block's sentences (*shuffled*) by +0.25 (ns): the graded facts, not the surface names or their order, carry most of the signal, consistent with a judge grading factual content over lexical form. Both are weak probes by construction (shuffling whole sentences preserves the facts within each sentence; masking only seed-class titles leaves the answer-bearing target names intact), so their small costs bound the surface-form contribution loosely rather than exactly. The arm-versus-raw contrasts also run over different survivor sets, so direct common-pair contrasts are the comparable reading: true − shuffled +0.30 [−0.10, +0.73] ($n = 30$) and true − masked +0.15 [−0.19, +0.44] ($n = 27$), both non-significant, point the same way while remaining imprecise. Finally, the effect is a **curation-depth gradient**: per set, true − raw is +0.95 [+0.37, +1.63] ($p = 0.018$) on the deep arcane set but only +0.08 [−0.46, +0.62] (ns) on the thin set. Where the corpus is rich the served block transfers real knowledge; where it is sparse there is little to transfer. This is the same gradient the live study shows, recovered independently through the controls.

7.5 Out-of-domain: the gate holds

Two instruments agree that grounding does not cost anything on the general questions a model already answers. The five-model judged arm (§6) graded the 60-question out-of-domain set 0–5 against frontier gold with a cross-family judge: every paired harness−bare delta's 95% interval includes zero: all questions −0.05 [−0.11, 0.00]; off-domain −0.09 [−0.25, +0.02]; and even the

Table 6: Negative-control contrasts, pooled arcane + thin, judged 0–5, paired mean difference with seeded bootstrap 95% CI and two-sided Wilcoxon p. Complete-case over surviving pairs; the control arms were not retried at the larger budget, so n varies by arm (§7). Read top to bottom: the serving path adds nothing (loom−true ≈ 0); the served block lifts quality (true−raw); the same-shaped placebo does not (irrelevant−raw ≈ 0); the content-specific residual is true−irrelevant.

Contrast	n	Δ	95% CI	Wilcoxon p
loom − true	32	−0.06	[−0.38, +0.25]	0.81
true − raw	32	+0.59	[+0.16, +1.06]	0.026
masked − raw	33	+0.42	[−0.09, +0.94]	0.15
shuffled − raw	36	+0.25	[−0.39, +0.89]	0.42
irrelevant − raw	28	+0.04	[−0.54, +0.57]	0.73
true − irrelevant	23	+0.35	[0.00, +0.78]	0.091
true − shuffled	30	+0.30	[−0.10, +0.73]	0.209
true − masked	27	+0.15	[−0.19, +0.44]	0.398
Per set (true − raw): arcane +0.95 [+0.37, +1.63], $p = 0.018$; thin +0.08 [−0.46, +0.62], ns.				

Table 7: Out-of-domain non-regression (five-model judged arm): paired harness−bare quality delta (judge 0–5 vs frontier gold), five models, seeded paired bootstrap 95% CI. Every interval includes zero. The worst case is off-domain questions where the confidence gate mis-fired and injected anyway; the last row is where it correctly skipped.

Subset	Mean Δ	95% CI	n pairs
All general questions	−0.05	[−0.11, 0.00]	300
adjacent	+0.01	[−0.02, 0.05]	100
in-domain-general	−0.06	[−0.14, 0.00]	100
off-domain	−0.09	[−0.25, 0.02]	100
Off-domain, gate mis-fired (worst)	−0.23	[−0.63, 0.05]	40
Off-domain, gate skipped (correct)	+0.00	n/a	60

pre-specified worst case, off-domain questions on which the gate mis-fired and injected anyway, −0.23 [−0.63, +0.05], still spanning zero (Table 7). The mechanism is the gate: when it correctly judged a query off-domain and skipped (60 of 100 off-domain gradings) the harnessed answer was identical to bare; the only degradation appeared when it over-fired, and it concentrated on the weakest model. Study 2 adds the equivalence reading the judged arm lacked: the general-set $\Delta = +0.05$ [0.00, +0.13] lies wholly inside the pre-specified ±0.25 TOST margin, so non-regression is established by CI-inclusion rather than inferred from a non-significant difference. On both instruments we report "no jaggedness detected" rather than "zero effect": the intervals are wide and bound the regression near zero, they do not prove its absence.

8 Analysis: Exposure, not Reasoning

The results across both studies say one thing in several registers: the uplift a graph-grounding layer delivers over a private corpus is overwhelmingly *exposure* (faithful delivery of curated facts the model was shown), not *reasoning* over injected structure. This section reads that claim off the instrument, then draws the two engineering consequences that reframe how such a system should be built and evaluated.

The headline is delivery fidelity, and the instrument is what reveals it. The ten-model sweep is the cleanest statement: raw uplift is huge and near-constant (+0.57 to +0.78), yet gain over copy is small and *uniformly negative* (−0.067 to −0.022). Read against a bare-model baseline every model looks transformed; read against the copy ceiling none exceeds what a verbatim copy of its own context already exposes, and they separate only by how much surfaced gold they drop. The number the field would report as the achievement (the uplift) is the number that carries the least information about the model; the number that discriminates (gain over copy) is invisible without the ceiling. That inversion is the whole case for reporting the ceiling as standard.

The live controls point to content, not form. The production controls take the same conclusion from headline to mechanism. The served block lifts judged quality (true − raw +0.59); a same-shaped, on-corpus but wrong block shows a near-zero point estimate (irrelevant − raw +0.04); the content-specific residual is +0.35; and masking names (+0.42) or shuffling sentences (+0.25) costs comparatively little. On these controls the graded facts appear to carry most of the effect, though the available arms do not isolate the contributions of names and order precisely. Three caveats keep this honest: the arms are complete-case over unequal survivor sets, the contrasts do not sum arithmetically (different survivors per arm), and several are non-significant with wide intervals. But the qualitative pattern (content-specific lift present, verified-irrelevant placebo near zero) is exactly what "exposure, faithfully delivered" predicts and what "any ontology-shaped text primes the model" does not. The placebo episode is the load-bearing methodological point: the effect only appeared clean once the irrelevant donor was *verified* seed-IRI-disjoint; a placebo assumed inert had leaked domain vocabulary and manufactured a spurious baseline.

Consequence 1: the copy ceiling is a regime detector. Compute it before you architect. The ceiling is not only a post-hoc control; it is a design-time diagnostic. Compute it on a representative question set *before* choosing an architecture. A near-1 ceiling (our setting: 0.964) means the answer's entity names are already exposed in the injected context; it does not by itself show that the relation needed to answer is present or understood, so the regime framing below is a heuristic, not a theorem. Where the ceiling is high the setting is a *serving regime*, in which retrieval already supplies most of the target coverage and generative re-encoding risks losing fidelity (though generation may still add selection, relevance and presentation), so the engineering target is faithful delivery and the metric is gain over copy. A low ceiling means the answer is *not* present as-is and must be composed across retrieved pieces: a *synthesis regime*, where GraphRAG-style machinery (community summaries, multi-hop traversal, GNN soft-prompts) can earn its keep. The essential discipline is that in a synthesis regime that machinery must be measured *above the ceiling*, not above a bare-model baseline: an uplift over the bare model in a high-ceiling setting is delivery masquerading as reasoning. Much reported graph-RAG gain is, we suspect, un-diagnosed serving-regime exposure; the ceiling is the cheap test that tells the two apart before a team builds the expensive apparatus.

Consequence 2: value concentrates at curation, and the instrument doubles as a promotion gate. If the model contributes delivery and the corpus contributes the answer, then the place to invest is upstream, at curation time, which is where the economics of a private corpus already put it, since the curation is amortised once and re-paid on no query. This gives the instrument a second use. An automated extraction candidate (for example a block proposed by OntoCast [13], analysed here purely as a third-party upstream producer) can be scored by the copy ceiling of the questions its block is meant to answer: a candidate whose block drives the ceiling toward 1 is answer-complete and worth promoting; one that does not is not yet ready. The same quantity that audits the serving layer thus grades the ingestion layer (answer-completeness of a block $\approx$ its copy ceiling), turning the control into a governance gate at the boundary where value is actually created. This gate is necessary, not sufficient: a block can inflate its copy ceiling by term-stuffing the answer names while remaining incoherent or wrong, so exposure coverage sits alongside the existing CI, consistency and human-review gates rather than replacing any of them.

The honest discovery arc. We did not set out to write this. We built the serving node first, saw large uplift, and worried we had merely reinvented GraphRAG with a nicer provenance story. The copy ceiling began as an adversarial question we put to our own result—*is any of this reasoning, or*

Table 8: The multivariate bar and this work's status. A deployable private-knowledge system must clear all four *together*; the axes are measured on different instruments, so no single number establishes the conjunction.

Axis	Instrument & evidence	Status
1. In-domain delivery fidelity	Copy ceiling / gain over copy: ten models -0.067 to -0.022; live loom$-$raw $+0.27$ pooled, $+0.79$ arcane	**Measured** (both studies)
2. Out-of-domain non-regression	Judged Δ vs frontier gold: five-model arm all CIs include zero (worst -0.23 $[-0.63, +0.05]$); live general $+0.05$ $[0.00, +0.13]$ inside ± 0.25 TOST margin	**Measured** (both studies)
3. Beats web search in-domain	Same model grounded in live web results vs curated scaffold, identical question set	Specified; not measured
4. Attribution precision	Per-answer generation + confidence already emitted; precision metric to close "faithful" → "attributable"	Specified; not measured

is it all exposure?—and the honest answer, uniformly negative gain over copy across ten models and a placebo whose point estimate collapses once verified irrelevant, is that it is exposure. For a private-knowledge product that is not a disappointment but the specification met: trustworthy because the source is vetted, attributable because the generation is named, and cheap because the curation is paid once. The contribution is the instrument that let us tell the difference, and the discipline of reporting it.

9 The Multivariate Bar

No single number is the deployable target. The bar introduced in §2 is conjunctive across four axes; Table 8 states each, the instrument that measures it, and this work's honest status against it. We shrink rather than inflate the scorecard: two axes are measured here, two are specified and not yet closed.

Axes 1 and 2 are the conjunction this paper establishes: excellent, attributable delivery where the corpus is authoritative, *and* no detected regression where it is not, measured on two different instruments so the pairing is not an artefact of one metric. Axis 3 (the right baseline is a web-grounded arm on the same questions, not the bare model) and axis 4 (a reported attribution-precision figure) are designed and unmeasured; we name them as open rather than imply the bar is cleared. The value of stating the full bar is that it prevents the in-domain number from being read as the whole claim: the failure mode the copy ceiling exists to prevent.

10 Limitations and Threats to Validity

- **Single corpus.** Both studies run on one curated ontology in one broad domain. The copy ceiling and its decomposition are corpus-general in construction, but the magnitudes (a 0.964 ceiling, a $+0.27$ live lift) are properties of this corpus and do not transfer as numbers.
- **Single-family judge.** Study 2's judge is a single GPT-family model (`openai/gpt-4.1`). It is cross-family from the Qwen candidate (avoiding self-preference) but is not a panel, is run without answer-order randomisation, and has no repeatability test across judge seeds; a judge idiosyncrasy would propagate. The gold references mitigate but do not eliminate this.
- **Complete-case control attrition.** The control arms were not retried at the larger budget and carry 36–50% empty attrition at 1536, so every control contrast is complete-case over unequal survivor sets. Because the attrition is condition-dependent (it rises with scaffold length and disorder), the survivors are not a random subsample and the complete-case contrasts are not safely causal; a uniform-budget rerun with every arm at the larger budget is the upgrade path. The arms do not sum arithmetically, several are non-significant with wide intervals

(shuffled−raw, masked−raw, true−irrelevant), and the attribution rests on the *pattern* across arms rather than any single contrast.

- **Small per-set n for controls.** Per-set control contrasts run on $n = 8$–21; the curation-depth gradient (arcane true−raw +0.95 significant, thin +0.08 ns) is suggestive, not definitively estimated, and the thin-set nulls are as consistent with low power as with no effect.
- **Lexical scorer for Study 1.** The ten-model gain-over-copy and in-domain recall use a lexical title matcher that under-counts paraphrase; absolute recall is a lower bound and gain over copy is framed as a lower-bound estimand. Only Study 2 uses the semantic judge.
- **Semantic fallback disabled throughout.** The HNSW semantic retrieval path was off in every reported run: its document-embedding recall (0.816) is below the 0.87 design floor (a red gate we report rather than relax), so the loom arm is lexical-retrieval only. A query-shaped embedding that clears the floor could change the engagement and thin-set numbers.
- **Single production model.** Study 2 holds the model at Qwen3.8-27B. The serving-path effect is isolated cleanly, but its magnitude under other backends is unmeasured; the ten-model sweep varies the model only for the offline scaffold, not the live node.
- **Three excluded questions.** Three questions empty at 4096 in both arms are dropped (complete-case $n = 114$). The exclusion is symmetric across arms but is a departure from intention-to-treat.
- **Two axes of the bar unmeasured.** The web-search baseline (axis 3) and attribution precision (axis 4) are specified and not measured here; the deployable claim is not yet fully closed.
- **The experiment a positive gain would demand.** Nothing here can license reasoning-over-structure, because every in-domain answer's target is verbatim in the scaffold and the decomposition finds essentially no beyond-exposure recovery ($n_{01} = 3$ of 11,360). The definitive test is a multi-hop design whose target is *not* verbatim in the injected context: over questions requiring composition of two graph edges, run arms that serve both premises, one premise removed, relation labels masked, matched irrelevant premises, and raw. Only an interaction there (recovery present with both premises, absent when either is removed) would separate reasoning over structure from parametric leakage or guessing. A positive gain over copy is the signal that would make this experiment worth running.

11 Conclusion

Grounding evaluations whose gold is derived from the injected corpus should **report a copy ceiling as standard.** It is a cheap, per-item, model-free quantity (the recall a verbatim copy of the shown context already achieves); the signed gain over copy it defines re-centres recall against exposure, and the exposure/recovery decomposition it anchors separates dropped exposed facts from beyond-exposure recovery, a distinction a bare-model baseline cannot make. Across ten models the gain is uniformly negative, and an exposure/recovery decomposition over 11,360 gold items measures the item-level claim directly: only 3 unexposed gold items are recovered and per-model context utilisation is 0.900–0.955, so recovery beyond exposure is negligible. A production paired study lifts judged quality by +0.27 pooled and +0.79 where curation is deep; and a four-arm control study is consistent with that lift being content-specific, the placebo's point estimate collapsing once verified seed-disjoint. The reusable method is the pair: *the ceiling plus a verified-irrelevant placebo*, computed before architecting to tell a serving regime (verbatim retrieval wins) from a synthesis regime (where heavier machinery must be measured above the ceiling, not above the bare model). And the economics follow the measurement: value concentrates upstream at curation, where the same instrument doubles as a promotion gate for extraction candidates. For a private corpus the goal was never a model that reasons its way to answers it never saw—it is a model that delivers vetted knowledge faithfully, attributably and cheaply, and an instrument honest enough to prove that is what it does.

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